F70LB Life Insurance Mathematics: Emerging cashflows and profit-testing

Heriot-Watt University, Edinburgh

2024-25 academic year

1 Before we begin: Some housekeeping

Welcome to this chapter of the F70LB course in which we look at Excel techniques, emerging cashflow techniques and profit-testing. Before we get started, you need to know how to use these notes and accompanying slides which will be discussed in class.

The notes and the slides are essentially the same. The slides are mostly a summary of the notes. However, there are (intentional!) gaps in these notes for exercises which we cover in class. The answers will be in the slides.

Sometimes the notes will contain extra detail that's not in the slides or is explained verbally in class. For example, the notes might include full details of a proof which is only sketched in class.

So what's the takeaway? You need both! You'll use these notes for revision, and you might find it helpful to bring an electronic or printed copy with you to class to annotate as we go along. Slides will be available after class. You won't generally need to refer back to slides unless you want to check the answer to an exercise.

Parts of the notes that are also in the slides are in blue font. This is to help you follow the notes in class, if you would like to.

I hope that's clear! Let's begin ...

Contents

1 Before we begin: Some housekeeping **1**

2 Introduction **3**
2.1 Exercise: Recap of our calculation methods . . . 3
2.2 Solution: Recap of our calculation methods . . . 3
2.3 Discussion: What's the problem with our traditional approach? . . . 4
2.4 Next steps . . . 5

3 Introducing profit-testing **6**

4 Profit criteria **9**
4.1 Introducing profit criteria . . . 9
4.2 The risk discount rate (RDR) . . . 10
4.3 Profit criteria . . . 11
4.4 Expected Profit at RDR . . . 11
4.5 Profit Margin . . . 12
4.6 Discounted Payback Period (DPP) . . . 12
4.7 Internal Rate of Return (IRR) . . . 12
4.8 Profit Measurement in Practice . . . 13
4.9 Practice calculating profit criteria . . . 13

5 Reserves **14**
5.1 Recap of reserves . . . 14
5.2 Reserves in PRO_t . . . 14
5.3 Example of profit testing reserves . . . 15

6 Actuarial bases **19**

7 Summing up **22**

2 Introduction

> . . . the end of all our exploring will be to arrive where we started and know the place for the first time . . . " T.S. Eliot, Little Gidding

The theory you have learned so far will underpin the applied skills that you use as an actuary. In this course, we are continuing to develop those tools into a real-world context through the use of Excel.

In this chapter of the course, we introduce what are called emerging cashflow techniques, and we apply them to something called profit testing.
To get started we'll need to step back a bit into the theory you've met before. (That's the reason for the quotation above.) By the end of this chapter, you'll see why!

Let's start with an exercise:

2.1 Exercise: Recap of our calculation methods

Consider a non-profit endowment assurance with the following contractual features:

- Term $n = 5$ years
- Age at issue $x = 60$
- Sum assured $S = £10,000$ paid at the end of year of survival or earlier death
- Initial expenses: £100
- Renewal expenses: 5% of premium per year.

Evaluate the annual premium P paid at the beginning of each year such that the expected profit is £50, using the traditional approach, with AM92 Ultimate mortality and 4% a year interest.

2.2 Solution: Recap of our calculation methods

We will cover the solution in class and it will be included in the slides. If you like, you can write the solution in this blank space before proceeding to the following page.

2.3 Discussion: What's the problem with our traditional approach?

We will discuss this in class. If you like, you can make notes in this blank space before proceeding to the following page.

2.4 Next steps

We need to take a step back! We can introduce maximum flexibility by looking at every single cashflow in every future time period separately. For example, for an annuity $\ddot{a}_x$ we could construct a spreadsheet with a column for each of the following:

- **time** $t = 0, 1, 2, 3, \ldots$
- a **payment** of £1 every year
- the **probability** of the payment being made $({}_tp_x)$
- the **discount** (v^t).

We can calculate $\ddot{a}_x$ in Excel using the functions SUM() or SUMPRODUCT() with our one of our formulas for $\ddot{a}_x$ from F70LA:

$$\ddot{a}_x = \sum_{t=0}^{\infty} {}_tp_x v^t 1$$

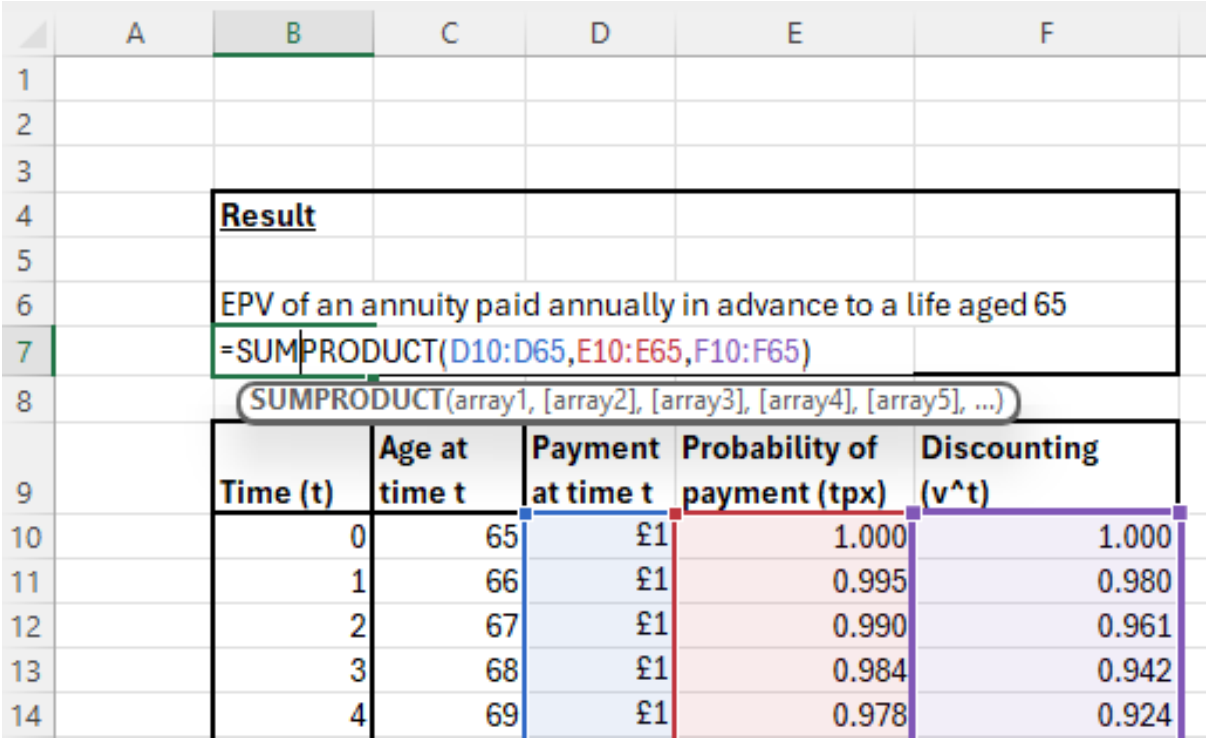


Figure 1: Calculating the EPV of an annuity in Excel

This gives us total freedom to control the mortality assumption and the amount of discounting. We can use any complicated mortality we like to calculate $({}_tp_x)$ for each $t = 0, 1, 2, 3, \ldots$ and we can use a yield curve (i.e., lots of different interest rates in different years). We can even change our payment column to anything we like! For example, payments that increase or which only start after a certain amount of time. We can model all sorts of complicated product features.

In section 3 *profit-testing* onwards we will use the power of splitting down cashflows year-by-year to take a more sophisticated approach to setting premiums and reserves.

3 Introducing profit-testing

We now turn to the main subject of this chapter: profit-testing. Profit-testing means the combination of three things:

1. projecting cashflows to each future time period
2. allowing for reserves
3. using profit criteria like DPP and IRR

We'll start with just the first. The first of these is broadly what we've been doing so far in this chapter: We've been breaking down our calculations into year-by-year cashflows, not just jumping straight to the expected present value. This breakdown gives us (and insurers!) a much richer insight into what is happening.

For profit-testing, we normally set out our cashflows in particular way, as follows.

Define CF_t to be the total expected cashflows emerging from time $t-1$ to t, accumulated to time t, given that the policy is in force at $t-1$.
Define the following elements that make up CF_t:

- P_t: Premium at $t-1$
- E_t: Expenses at $t-1$
- i_t: Investment return $t-1 \to t$
- q_{x+t-1}: Probability of death
- $p_{x+t-1} = 1 - q_{x+t-1}$
- D_t: Benefit on death
- S_t: Benefit on survival to t.

Then:

$$CF_t = P_t - E_t + i_t(P_t - E_t) - q_{x+t-1}D_t - p_{x+t-1}S_t \tag{1}$$

This can be set out in a table or a spreadsheet using each of these as a column heading.
Let's look at our example from section 2.1 again.

Non-profit endowment assurance, term $n = 5$ years, age at issue $x = 60$, sum assured $S = £10,000$ paid at the end of year of maturity or earlier death, initial expenses £100, renewal expenses 5% of premium per year, AM92 Ultimate mortality and 4% a year interest. We used our F70LA approach to calculate the premium needed to get a profit of £50 as follows:

$$\begin{aligned} 0.95P\ddot{a}_{60:\overline{5}|} &= 10,000A_{60:\overline{5}|} + 50 + 100 \\ P &= £1943.30 \end{aligned}$$

Now, let's check that if we set out our calcs as a profit test, then we'd get the same answer!

Question: We want to create a table that calculates CF_t, but that depends on our premium. The premium is the unknown we're trying to find, so what's the easiest way to handle this in Excel?

Answer: It's easiest to enter a dummy value and then use goalseek in Excel to find the correct value. We can use a named range to make our calculations nice and neat! Let's guess £1500 for our premium for now.

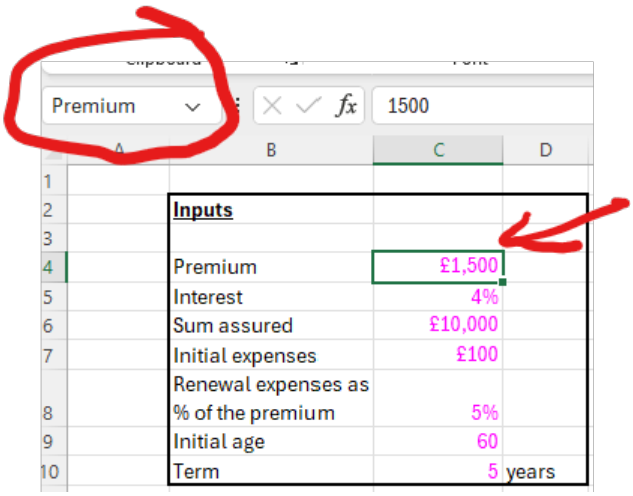


Figure 2: Start with a guess of the premium!

Now let's set out our calculation of

$$CF_t = P_t - E_t + i_t(P_t - E_t) - q_{x+t-1}D_t - p_{x+t-1}S_t$$

Calculations

Time at year end (t)	Age at time t	P(t)	E(t)	i(t)*(P(t) - E(t))	q(x+t-1)	p(x+t-1)	D(t)	S(t)	CF(t)
1	61	1500	175.00	53.00	0.008	0.992	10000	0	1,297.78
2	62	1500	75.00	57.00	0.009	0.991	10000	0	1,391.91
3	63	1500	75.00	57.00	0.010	0.990	10000	0	1,380.88
4	64	1500	75.00	57.00	0.011	0.989	10000	0	1,368.56
5	65	1500	75.00	57.00	0.013	0.987	10000	10000	-8,518.00

Figure 3: Calculation of CF_t

Note that:

- $E_1 = £100 + 4\%P_t$ and $E_2 = 4\%P_t$
- q_{x+t-1} and p_{x+t-1} come from the AM92 table
- We'll need to change P_t later to get a profit of £50

We interpret CF_t as follows:

- CF_t gives us the total cashflow occuring each year *given that the policy is still in force at the start of the year*, accumulated to the end of the year.
- It is the 'surplus' or 'deficit' from that year.

Figure 4 illustrates how we can calculate the EPV of profit in Excel using SUMPRODUCT(). We discount back using v^t and allow for the probability that each CF_t occurs by using ${}_{t-1}p_x$. Note that we can calculate ${}_{t-1}p_x$ using p_{x+t-1} by starting from ${}_0p_x = 1$ with the relationship ${}_tp_x = ({}_{t-1}p_x)(p_{x+t-1})$.

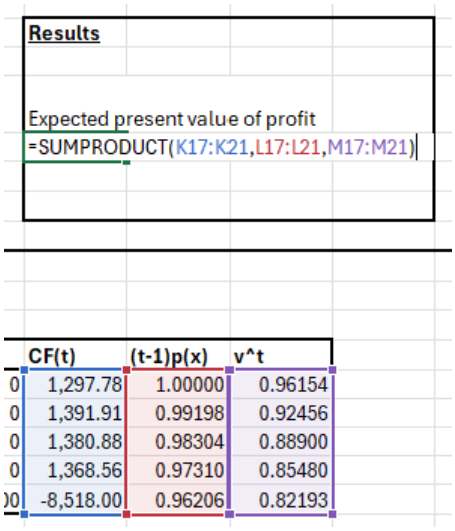


Figure 4: Calculation of EPV of profit in Excel using CF_t

We now use goalseek in Excel to find the premium P such that

$$\begin{aligned} £50 &= \text{Expected present value of profit} \\ &= \sum_{t=1}^{5} {}_{t-1}p_x v^t CF_t \end{aligned}$$

The answer we get is $P = £1943$ This is exactly what we expected, so what have we gained?

- We've checked this method makes sense!
- We can see when surplus/deficit emerges

CF(t)
1,735.70
1,829.83
1,818.80
1,806.48
-8,080.08

Figure 5: CF_t each year

But we can do more! In the following sections, we will look at an allowance for reserves, for different assumptions and for profit tests themselves.

4 Profit criteria

4.1 Introducing profit criteria

We are now going to look at different ways to allow for profit in life insurance.

Before we proceed, let's do a quick quiz. **Quiz:** Which of the following is **not** a way to include an allowance for profit in a term assurance premium?

1. Assume higher-than-expected mortality in the premium basis
2. Project each year's cashflows individually and then use techniques like Internal Rate of Return and Discounted Payback Period
3. Assume higher-than-expected interest in the premium basis
4. Include an explicit allowance for profit in expenses

We will cover the solution in class and it will be included in the slides. If you like, you can write the solution in this blank space before proceeding to the following page.

Before we look at profit criteria, it is convenient to introduce some notation:

- The **profit vector** is $(PRO_1, PRO_2, \ldots, PRO_n)$ where PRO_t is the profit made in the year $t-1$ to t per policy in force at time $t-1$, after allowance for setting up reserves.
- The **profit signature** at time t is the profit made in year $t-1 \to t$, accumulated to time t, **per policy sold at time zero**, after allowance for setting up reserves. Denote it σ_t, then: $\sigma_t = {}_{t-1}p_x PRO_t$

PRO_t **is just** CF_t **plus an allowance for reserves.** We will explain how to calculate PRO_t in a later section. For now, you can assume $PRO_t = CF_t$.

(If there are no reserves then we will always have $PRO_t = CF_t$)

4.2 The risk discount rate (RDR)

When we calculated the expected present value of profit above

$$£50 = \sum_{t=1}^{5} {}_{t-1}p_x v^t CF_t \tag{2}$$

$$= \sum_{t=1}^{5} v^t \sigma_t \tag{3}$$

we used the same interest rate as in the premium (and reserving) basis.

And that gave us the same result as if we had used the traditional actuarial EPV functions at 4%. This is hardly a surprise. However, without exceptional complexity, it is the *only* answer to be got from the traditional approach.

With profit-testing we can be much more flexible. From the capital provider's point of view, CF_t represents their expected future cashflows, if they invest in the insurer writing this new business. Recall from Financial Mathematics:An investor will appraise investment opportunities using a variety of techniques. **Discussion:** What discount rate might the investor use in (4) above, to discount the profit signature?

We will cover the solution in class and it will be included in the slides. If you like, you can write the solution in this blank space before proceeding to the following page.

Definition: The *Risk Discount Rate (RDR)* is a discount rate chosen by the investor for deciding whether of not to invest. The investor can discount the profit signature at their chosen RDR, and then apply one or more *profit criteria* to help them decide. We will usually have more two or more discount rates we are interested in:

- i_r, the RDR
- i_p, the interest rate in the actuary's pricing and/or valuation basis.

4.3 Profit criteria

Four informative profit criteria are:

- Expected profit at RDR
- Profit margin
- Discounted payback period
- Internal rate of return.

We discuss each of these in turn.

4.4 Expected Profit at RDR

The expected profit at the RDR is:

$$\sum_{t=1}^{5} v^t \sigma_t \tag{4}$$

This is the same as that in the traditional approach (£50) if $i_r = i_p$. **Question:** Why is it very unlikely that $i_r = i_p$?

We will cover the solution in class and it will be included in the slides. If you like, you can write the solution in this blank space before proceeding to the following page.

In our example, the expected at profit at $i = 4\%$ was £50. The expected profit at $i_r = 10\%$ would be £33.56, which is less than £50 as expected because $i_r > i_p$.

If the insurer's profit target was £25 per policy then it would pass this test at $i_r = 10\%$ since £33.56 >£25

Let's look at some more tests.

4.5 Profit Margin

The profit margin is the profit expressed as a proportion of the EPV of the premiums, calculated using the RDR. So it is the share of the premiums that the insurer gets as profit.

$$\text{Profit Margin} = \frac{\text{EPV Profit @ RDR}}{\text{EPV of Premium @ RDR}}$$

Where:

$$\text{EPV Premium @ RDR} = \sum_{t=1}^{n} {}_{t-1}p_x v^{t-1} P_t$$

The profit criterion may then be a profit margin of $X\%$. For example, if we require a profit margin of 1% of premiums, our example policy would fail since the EPV of the premiums at 10% is £7972, so the profit margin is only 0.4%.

4.6 Discounted Payback Period (DPP)

DPP is the time taken to repay the new business strain with interest at the RDR. It is the smallest time t such that:

$$\sum_{s \le t} (1 + i_r)^{-s} \sigma_s \ge 0$$

The faster the strain is repaid (i.e. the smaller t), the sooner the insurer can invest in a new project. The insurer may set a target, for example that the DPP should be not more than 5 years.

Note that DPP only makes sense when we have one or more periods of net negative cashflows at the start followed by net positive cashflows.

Example: Suppose we have the profit signature σ_t shown in the second column below. Then in the third column we add up all the discounted cashflows up to time t.

t	σ_t	$\sum_{s \le t}(1+i_r)^{-s}\sigma_s$
1	−69.45	−63.14
2	34.01	−35.03
3	33.70	−9.71
4	33.36	13.08
5	32.99	33.56

The DPP is 4 years, since that is the first time that EPV turns positive, and less than half of the strain has been repaid by year 2.

4.7 Internal Rate of Return (IRR)

The Internal Rate of Return (IRR) is the rate of interest at which the EPV of future profits is zero.

It cannot always be used since there may be no solutions or more than one solution to the above equation.

$$\text{IRR} = i \text{ such that } \sum_t \sigma_t (1+i)^{-t} = 0$$

A profit criterion may be, for example, that the IRR is at least as great as the RDR, which means that the EPV of profits at the RDR is positive. Our example policy passes this test. Note that DPP and IRR have widely been used for evaluating profits from investment projects in corporate finance.

4.8 Profit Measurement in Practice

When deciding on the appropriate price which meets our profit criteria, we follow this procedure:

- Evaluate the four profit measures when designing a new product or setting premium rates.
- Adjust premiums/benefits until the profit criteria are passed.
- We can't ignore marketing considerations: must ensure that we are able to sell the contract (e.g. competitors' premiums).

Next, we will look at how to allow for reserves.

4.9 Practice calculating profit criteria

Exercise: Let's do a quick exercise. Assume $i_r = 10\%$ and $i = 4\%$. Use the following information to calculate all four profit criteria.

A	B	C	D	E	F	G	H	I	J
t	CF(t)	(t-1)p(x)	(t-1)p(x) CF(t)	PRO(t)	sigma(t)	v^t (at 4%)	v^t (at 10%)	Premium	v^t (at 25%)
1	5	1	5	-4.88	-4.88	0.96	0.91	1	0.80
2	5	0.99	4.95	3	2.97	0.92	0.83	1	0.64
3	-9	0.98	-8.82	4	3.92	0.89	0.75	1	0.51

	Sumproduct of columns	
1	D, G	1.54
2	D, H	2.01
3	F, G	1.54
4	F, H	0.96
5	H, I	2.49
6	F, J	0.00

Figure 6: Profit criteria exercise

We will discuss this in class. If you like, you can make notes in this blank space before proceeding to the following page.

5 Reserves

In this section, we will look at how to allow for reserves in profit-testing. We start with a reminder of what reserves are and how they might be calculated. Then we look at how to calculate the profit vector and how that affects our answers.

5.1 Recap of reserves

Let's start with a quick recap of reserves:

- A reserve is the amount of assets held by the insurer in respect of a group of policies
- The reserve is needed to:
  - Cover timing difference between outgo and income
  - Demonstrate solvency
- The reserve for a group of policies needs to be at least as big as the liability for those policies
- The liability might be calculated as a policy value

$$\text{Policy value} = \text{EPV(outgo minus income for the policy)}$$

5.2 Reserves in PRO_t

Let us consider PRO_t. Define V_t to be the reserve held per policy in force at time t.

- At time $t-1$, reserve V_t is held.
- During the year, interest of iV_{t-1} is earned. (Note: i is probably much lower than the RDR.)
- At the end of the year, a reserve of V_t must be set up for each policy then in force, but the reserve V_{t-1} is available to offset this.
- These items must be added to the 'actual' cashflow CF_t.

Hence:

$$\begin{aligned} PRO_t &= CF_t + \text{interest on reserves} - \text{increase in reserves} \\ &= CF_t + iV_{t-1} - (p_{x+t-1}\, V_t - V_{t-1}) \end{aligned}$$

The term $(p_{x+t-1}\, V_t - V_{t-1})$ is the increase in reserves and is often denoted ΔV_t.

If it is positive then the insurer must inject capital (which has a cost). If it is negative, capital is released back to the insurer and this contributes to surplus.

5.3 Example of profit testing reserves

Let's return to our example of a 5-year endowment assurance. We assume cashflows and reserves earn $i = 4\%$ interest each year. Recall that (with no allowance for reserves) if the RDR is also 4% then the EPV of profit is £50.

Suppose for simplicity that reserves are £0, £2000, £4000, £6000, £8000 at times $t = 0, 1, 2, 3, 4$ respectively.

These reserves are higher than the gross premium policy values at times $t = 0, 1, 2, 3, 4$ calculated using $i = 4\%$.

Exercise: If you have time, you could check that the gross premium policy values are £1697, £3628, £5649, £7769 at times $t = 1, 2, 3, 4$. The gross premium policy value at time 0 would be $-$£50, so we take it to be £0.

Question: Before we proceed, consider, how do you expect the allowance for reserves to affect our answers? Will EPV of profit be lower or higher? Why?

In Figure 7, we continue our example and calculate PRO_t in Excel using the relationship $PRO_t = CF_t + iV_{t-1} - (p_{x+t-1}\ V_t - V_{t-1})$.

			Interest on reserves	Increase in reserves DELTA(V(t))	
CF(t)	V(t)	V(t-1)	iV(t-1)	p(x+t-1)V(t) - V(t-1)	PRO(t)
1,736	2,000	0	0	1,984	-248
1,830	4,000	2,000	80	1,964	-54
1,819	6,000	4,000	160	1,939	39
1,806	8,000	6,000	240	1,909	137
-8,080	0	8,000	320	-8,000	240

Figure 7: Allowing for reserves in PRO_t

Question: Compare PRO_t and CF_t. What do you notice?
We will discuss this in class and it will be included in the slides. If you like, you can write the solution in this blank space before proceeding to the following page.

Definition: Negative profits in the early years of a policy are called *New Business Strain*. We expect new business strain in most policies because:

- We need to create reserves at the start
- There are often big expenses at the start, especially underwriting and commission.

CF(t)	PRO(t)
1,736	-248
1,830	-54
1,819	39
1,806	137
-8,080	240

Figure 8: CF_t vs PRO_t, impact of reserves

Question: What do you expect the EPV of profit to be in each of these cases, after allowing for reserves? Remember that $i = 4\%$ on reserves and cashflows. More or less than £50?

- RDR = 4% (unlikely)
- RDR = 9%
- RDR = 2% (very unlikely!)

We will cover the solution in class and it will be included in the slides. If you like, you can write the solution in this blank space before proceeding to the following page.

Important result: Suppose the interest rate received on reserves and cashflows is the same as the RDR. Then the size of reserves makes no difference to the expected present value of profit. This is a general result, not just true in our example.

Proof: By convention, we assume that $V_0 = 0$ since it is the reserve immediately before the policy starts, i.e. there is no liability yet because the policy hasn't begun. We also assume that $V_n = 0$, where n is the term of the policy, since at the end of the policy there is nothing to reserve for. (We assume that V_n is the reserve immediately after any maturity benefit is paid.)

The EPV of profits is:

$$\sum_{t=1}^{n}(1+i_r)^{-t}\ {}_{t-1}p_x\ PRO_t$$

where i_r is the RDR.
If i_r is the same as the experience basis interest rate i, this is:

$$= \sum_{t=1}^{n}\left((1+i)^{-t}\ {}_{t-1}p_x \times \left[CF(t) + {}_{t-1}V(1+i) - p_{x+t-1}\ {}_tV\right]\right).$$

This is:

$$= \sum_{t=1}^{n}(1+i)^{-t}\ {}_{t-1}p_x\ CF_t + \sum_{t=1}^{n}(1+i)^{-t+1}\ {}_{t-1}p_x\ V_{t-1} - \sum_{t=1}^{n}(1+i)^{-t}\ {}_tp_x\ V_t.$$

Let $s = t-1$ in the middle sum:

$$\begin{aligned}
&= \sum_{t=1}^{n}(1+i)^{-t}\ {}_{t-1}p_x\ CF_t + \sum_{s=0}^{n-1}(1+i)^{-s}\ {}_sp_x\ V_s - \sum_{t=1}^{n}(1+i)^{-t}\ {}_tp_x\ V_t \\
&= \sum_{t=1}^{n}(1+i)^{-t}\ {}_{t-1}p_x\ CF_t + V_0 - (1+i)^{-n}V_n \\
&= \sum_{t=1}^{n}(1+i)^{-t}\ {}_{t-1}p_x\ CF_t \quad \text{since} \quad V_0 = 0 = V_n
\end{aligned}$$

This final line does not depend on the reserves, hence the result.

6 Actuarial bases

Recall that an actuarial basis is a set of **assumptions** used to carry out one or more actuarial calculations. The assumptions usually include interest, mortality and expenses, but may also include items such as bonus, inflation, surrenders and morbidity.

Different bases may be used for *different* purposes. For example:

- The pricing actuary may choose a cautious set of assumptions when setting premiums — the pricing or premium basis.
- The valuation actuary will have to use a basis that satisfies the regulator— the valuation basis.
- The finance director wants to estimate future profits using a realistic estimate of the future — the experience basis.

We sometimes use the term *experience basis* in two different ways. One way is forward-looking (as above) and means a realistic estimate. The other way is backward-looking and means what has actually happened, i.e. actual experience.

In the equation for PRO_t, the premiums would be calculated on the premium basis, the policy values on the valuation basis and all other elements of calculating PRO_t on the experience basis — *and these could all be different.*

Also important are *scenario testing* or *sensitivity analysis.*

Scenario testing and sensitivity analysis are ways to ask what happens if the future turns out differently to our assumptions? By varying the experience basis, the impact on profits can be seen directly.

Sensitivity analysis usually means looking at changes in the basis one assumption at a time. This tells us how sensitive our results are to each of the assumptions we have made.

Scenario analysis means changing groups of assumptions together in some meaningful way. For example, we might look at the scenario of a cost of living crisis. Nominal interest rates might increase, lapses and surrenders might increase and new business might fall.

In general, we may ask what is the effect on profit of varying any element of any of the three bases: premium, valuation or experience? For the premium or experience bases, it is usually intuitively clear. For example, consider changes in the experience basis, as we would do in a sensitivity analysis.

Question: Effect of experience basis on profit?

- Higher interest ⇒ Higher/Lower profit?
- Higher expenses ⇒ Higher/Lower profit?
- High mortality:
  1. Term assurance ⇒ Higher/Lower profit
  2. Annuity ⇒ Higher/Lower profit

We will discuss this in class. If you like, you can make notes in this blank space before proceeding to the following page.

Question: Effect of premium basis on profit?

- Higher interest ⇒ Higher/Lower profit?
- Higher expenses ⇒ Higher/Lower profit?
- High mortality:
  1. Term assurance ⇒ Higher/Lower profit
  2. Annuity ⇒ Higher/Lower profit

We will discuss this in class. If you like, you can make notes in this blank space before proceeding to the following page.

Question: Effect of valuation basis on profit?

We know that higher reserves means lower profit so long as RDR$> i$, which it will be.

- Higher interest $\Rightarrow$ Higher/Lower profit?
- Higher expenses $\Rightarrow$ Higher/Lower profit?
- High mortality:
  1. Term assurance $\Rightarrow$ Higher/Lower profit
  2. Annuity $\Rightarrow$ Higher/Lower profit

We will discuss this in class. If you like, you can make notes in this blank space before proceeding to the following page.

7 Summing up

Key skills

On completion of this chapter of the course, you should be able to:

- Explain the purpose and benefits of profit-testing
- Calculate EPVs in Excel using SUMPRODUCT(), goalseek and other tools
- Use terminology from this chapter, such as Risk Discount Rate
- Calculate CF_t, PRO_t and $_t$
- Calculate profit criteria and draw conclusions from them
- Explain the effect of reserves and different actuarial bases

What next?

In the next section of the course, we look at unit-linked contracts. These are a modern product which rely on emerging cashflow techniques.